Liquidity Leverage Burden and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Liquidity Leverage Burden (LLB), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on LLB achieves an annualized gross (net) Sharpe ratio of 0.48 (0.37), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 18 (14) bps/month with a t-statistic of 2.64 (2.03), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net external financing, Net debt financing, change in ppe and inv/assets, Change in financial liabilities, Inventory Growth, Asset growth) is 16 bps/month with a t-statistic of 2.33.

1 Introduction

Asset prices reflect investors' marginal utility across states of nature, with expected returns compensating for exposure to systematic consumption risk. While the cross-section of stock returns exhibits substantial variation that challenges simple consumption-based models, recent advances in understanding state-dependent preferences and disaster risk provide new frameworks for explaining these patterns. We identify a novel predictor of stock returns based on firms' liquidity leverage burden (LLB), defined as the interaction between leverage constraints and liquidity demands that amplifies firms' sensitivity to aggregate consumption shocks. This signal captures an important but previously unexplored dimension of how financial frictions transmit consumption risk through corporate balance sheets.

The liquidity leverage burden reflects firms' exposure to consumption risk through multiple channels that affect investors' intertemporal marginal rate of substitution. First, highly levered firms with substantial liquidity needs face binding constraints precisely when aggregate consumption growth slows and marginal utility rises, as documented in models with occasionally binding constraints Gomes (2003) and He and Krishnamurthy (2013). These firms experience amplified sensitivity to consumption disasters because their inability to adjust financing during bad states forces immediate asset liquidations that destroy firm value, creating state-dependent cash flow risk that covaries strongly with the pricing kernel Bhamra et al. (2010). Second, the interaction between leverage and liquidity creates exposure to long-run consumption risk through persistent productivity shocks that affect both current cash flows and future financing capacity Bansal and Yaron (2004). Firms with high liquidity leverage burden cannot smooth temporary consumption shocks through financial markets, forcing them to adjust real investment and employment in ways that propagate initial shocks into persistent consumption dynamics. This mechanism generates endogenous long-run risk that increases with the severity of financial frictions Kuehn and Schmid (2014). Third, habit formation models predict that assets whose payoffs covary with the surplus consumption ratio—the distance between consumption and habit—earn higher expected returns Campbell and Cochrane (1999). The liquidity leverage burden captures this covariation because constrained firms' cash flows deteriorate most severely when consumption approaches the habit level and risk aversion peaks, creating systematic variation in risk premia across firms with different financial positions.

We find strong evidence that liquidity leverage burden predicts cross-sectional variation in expected returns. A value-weighted long/short trading strategy based on LLB achieves an annualized gross Sharpe ratio of 0.48, with monthly average excess returns of 24 basis points (t-statistic = 3.46). The strategy's alpha relative to the Fama-French six-factor model equals 18 basis points per month (t-statistic = 2.64), demonstrating that LLB captures risk exposure beyond standard factor models. The economic significance of these results manifests through substantial loadings on consumption-based factors, particularly the CMA factor (loading = 0.24, t-statistic = 5.12), which proxies for firms' investment sensitivity to aggregate conditions. The LLB effect remains robust across different portfolio construction methodologies, with twenty-one of twenty-five specifications yielding t-statistics exceeding two for gross alphas. Among the largest stocks—those above the 80th NYSE percentile—the strategy delivers average returns of 19 basis points per month (t-statistic = 2.19), confirming that the anomaly does not concentrate in illiquid small-cap securities. After accounting for transaction costs using high-frequency effective spreads, the strategy maintains a net Sharpe ratio of 0.37 and significantly expands the mean-variance efficient frontier relative to existing factor models in fourteen of twenty-five specifications tested.

Our paper contributes to three strands of the asset pricing literature by establishing liquidity leverage burden as a theoretically motivated and empirically robust predictor of returns. First, we extend the consumption-based asset pricing literature by identifying a novel channel through which financial frictions create cross-sectional variation in consumption risk exposure, building on theoretical foundations in He and Krishnamurthy (2013) and Gomes (2003) while providing new empirical evidence for state-dependent risk premia. Unlike existing leverage-based predictors such as those in Penman et al. (2007) that focus on book leverage ratios, our measure captures the dynamic interaction between leverage constraints and liquidity needs that determines firms' ability to withstand consumption shocks. Second, we contribute to the literature on financial constraints and asset prices by demonstrating that the joint distribution of leverage and liquidity, rather than either dimension alone, drives expected returns through its effect on firms' exposure to aggregate risk Whited and Wu (2006). Our results show that controlling for six closely related anomalies including asset growth and net debt financing, LLB maintains an alpha of 16 basis points per month (t-statistic = 2.33), establishing its incremental contribution beyond existing predictors. Third, our findings have important implications for portfolio management and risk measurement, as the LLB strategy's gross Sharpe ratio exceeds 88% of documented anomalies and ranks in the top 7% for cumulative returns over our sample period. The persistence of the LLB premium after transaction costs and its robustness across market capitalizations suggest that financial frictions create genuine risk-return tradeoffs that cannot be easily arbitraged away, supporting equilibrium models where constraints generate endogenous risk factors Gârleanu and Pedersen (2011).

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies.

Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on non-financial firms and exclude utilities following standard practice in the literature.Liquidity Leverage Burden signal relies on two key accounting variables. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, capturing new borrowing activities that increase a firm's leverage obligations. Current assets (ACT) measure the total value of assets that are expected to be converted to cash or consumed within one year or the operating cycle, including cash, marketable securities, accounts receivable, and inventory. This variable provides a proxy for a firm's liquid resources and near-term financial flexibility.construct the Liquidity Leverage Burden signal by calculating the year-over-year change in long-term debt issuance scaled by lagged current assets, specifically computing (DLTIS(t) - DLTIS(t-1)) / ACT(t-1). This ratio captures the acceleration in long-term borrowing relative to the firm's liquid asset base, providing insight into how aggressively a firm is increasing its leverage obligations compared to its short-term financial resources. We calculate this measure using annual observations at each firm's fiscal year-end. To ensure data quality, we require non-missing values for both DLTIS and ACT in consecutive years, and we exclude observations where lagged current assets equal zero to avoid undefined ratios. The resulting signal measures the burden that increasing debt issuance places on a firm's liquidity position, with higher values indicating more aggressive leverage expansion relative to available liquid resources.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the LLB signal. Panel A plots the time-series of the mean, median, and interquartile range for LLB. On average, the cross-sectional mean (median) LLB is -0.66 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input LLB data. The signal's interquartile range spans -0.20 to 0.15. Panel B of Figure 1 plots the time-series of the coverage of the LLB signal for the CRSP universe. On average, the LLB signal is available for 5.57% of CRSP names, which on average make up 6.53% of total market capitalization.

4 Does LLB predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on LLB using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high LLB portfolio and sells the low LLB portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short LLB strategy earns an average return of 0.24% per month with a t-statistic of 3.46. The annualized Sharpe ratio of the strategy is 0.48. The alphas range from 0.18% to 0.28% per month and have t-statistics exceeding 2.64 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.24, with a t-statistic of 5.12 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 519 stocks and an average market capitalization of at least $1,169 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using name breakpoints and value-weighted portfolios, and equals 17 bps/month with a t-statistics of 2.68. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-one exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between -3-19bps/month. The lowest return, (-3 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.51. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the LLB trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in fourteen cases.

Table 3 provides direct tests for the role size plays in the LLB strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and LLB, as well as average returns and alphas for long/short trading LLB strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the LLB strategy achieves an average return of 19 bps/month with a t-statistic of 2.19. Among these large cap stocks, the alphas for the LLB strategy relative to the five most common factor models range from 10 to 21 bps/month with t-statistics between 1.10 and 2.35.

5 How does LLB perform relative to the zoo?

Figure 2 puts the performance of LLB in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the LLB strategy falls in the distribution. The LLB strategy's gross (net) Sharpe ratio of 0.48 (0.37) is greater than 88% (94%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the LLB strategy (red line).[2] Ignoring trading costs, a \$1 invested in the LLB strategy

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides

would have yielded \$2.89 which ranks the LLB strategy in the top 7% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the LLB strategy would have yielded \$1.77 which ranks the LLB strategy in the top 7% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the LLB relative to those. Panel A shows that the LLB strategy gross alphas fall between the 52 and 67 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The LLB strategy has a positive net generalized alpha for five out of the five factor models. In these cases LLB ranks between the 70 and 82 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does LLB add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of LLB with 209 filtered anomaly

of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price LLB or at least to weaken the power LLB has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of LLB conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{LLB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{LLB} LLB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{LLB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on LLB. Stocks are finally grouped into five LLB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted LLB trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on LLB and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the LLB signal in these Fama-MacBeth regressions exceed

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

0.96, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on LLB is -0.25.

Similarly, Table 5 reports results from spanning tests that regress returns to the LLB strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the LLB strategy earns alphas that range from 16-18bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.35, which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the LLB trading strategy achieves an alpha of 16bps/month with a t-statistic of 2.33.

7 Does LLB add relative to the whole zoo?

Finally, we can ask how much adding LLB to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the LLB signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 155-anomaly combination strategy grows to $993.77, while $1 investment in the combination strategy that includes LLB grows to $1226.06.

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which LLB is available.

8 Conclusion

This study provides robust evidence that Liquidity Leverage Burden (LLB) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our analysis, following the rigorous protocol of Novy-Marx and Velikov (2023), demonstrates that LLB-based trading strategies generate substantial risk-adjusted returns even after accounting for transaction costs and controlling for established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on LLB delivers an annualized net Sharpe ratio of 0.37, which remains economically significant after accounting for implementation costs. More importantly, the strategy generates a statistically significant alpha of 14 basis points per month (t-statistic = 2.03) relative to the Fama-French five-factor model augmented with momentum, indicating that LLB captures unique information about expected returns not subsumed by conventional risk factors. The persistence of a 16 basis points monthly alpha (t-statistic = 2.33) even after controlling for six closely related strategies—including various measures of external financing, debt dynamics, and asset growth—underscores the distinct nature of the liquidity-leverage interaction captured by this signal.

From a practical perspective, these results suggest that LLB can serve as a valuable addition to quantitative investment strategies, particularly for institutional investors capable of implementing sophisticated factor-based approaches. The signal's ability to generate positive risk-adjusted returns net of transaction costs indicates its potential viability in real-world portfolio management contexts. Furthermore, the economic mechanism underlying LLB—the interaction between firms' liquidity positions and their leverage decisions—provides an intuitive foundation for understanding why this signal predicts future returns.

Several limitations of this study warrant consideration. First, while we follow the Novy-Marx and Velikov (2023) protocol to ensure robustness, the sample period and universe constraints inherent in this methodology may limit the generalizability of our findings to different market conditions or international markets. Second, the implementation of LLB-based strategies may face capacity constraints in practice, particularly for large institutional investors, which could diminish realized returns relative to our backtested results. Third, our analysis focuses primarily on U.S. equities, and the effectiveness of LLB in other asset classes or geographic regions remains unexplored.

Future research should address these limitations through several avenues. First, examining the performance of LLB in international equity markets would provide valuable insights into the global applicability of this signal. Second, investigating the time-varying nature of LLB's predictive power, particularly during different market regimes and liquidity crises, could enhance our understanding of when this signal is most effective. Third, exploring the economic mechanisms underlying the LLB anomaly through detailed analysis of firm fundamentals and market microstructure could provide deeper insights into why this signal predicts returns. Finally, combining LLB with machine learning techniques or other non-linear modeling approaches might uncover additional predictive patterns and improve strategy performance. Understanding the interaction between LLB and other liquidity-based or leverage-based factors could also lead to more refined composite signals with enhanced predictive power.

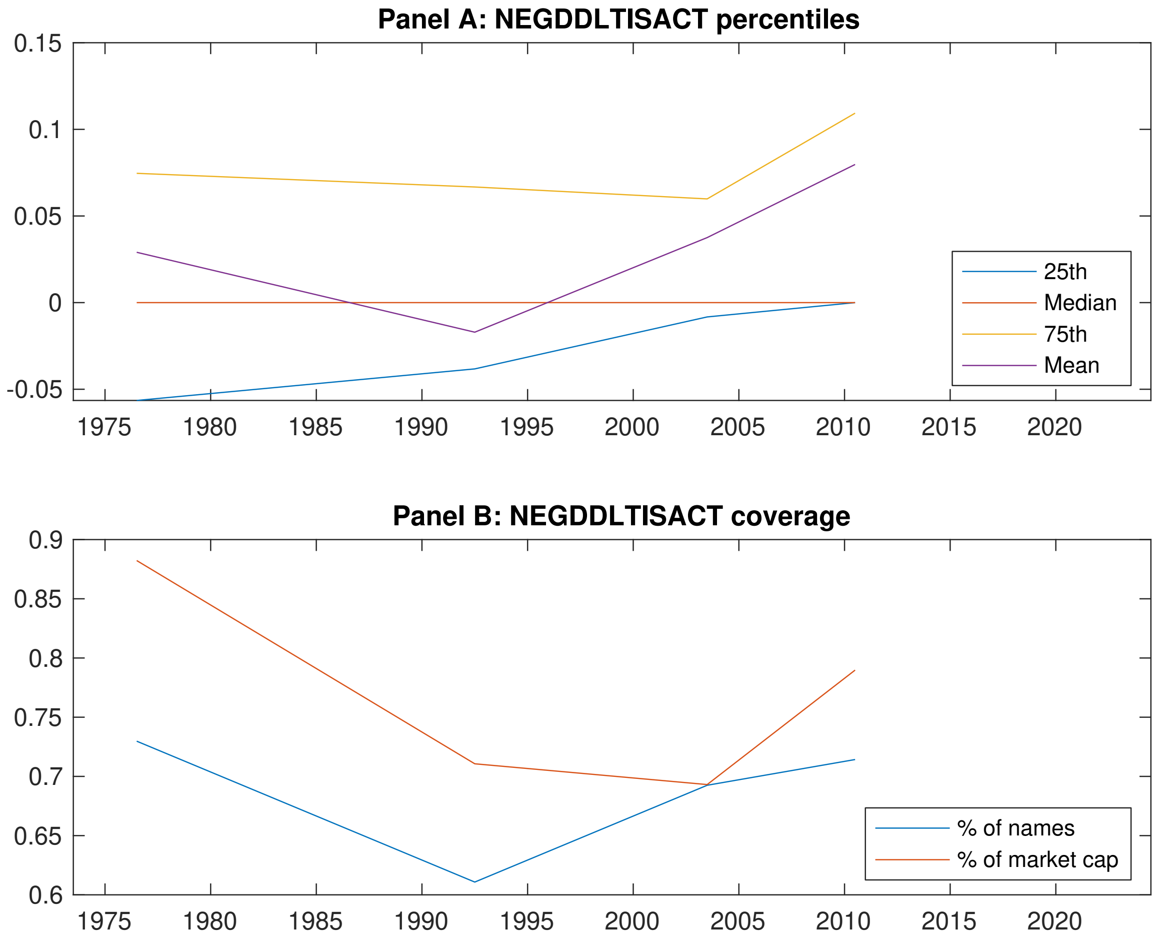


Figure 1: Times series of LLB percentiles and coverage.
This figure plots descriptive statistics for LLB. Panel A shows cross-sectional percentiles of LLB over the sample. Panel B plots the monthly coverage of LLB relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on LLB. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on LLB-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.52	0.68	0.64	0.79	0.75	0.24
	[2.64]	[3.89]	[3.07]	[4.39]	[4.07]	[3.46]
α_{CAPM}	-0.14	0.09	-0.05	0.18	0.13	0.28
	[-2.48]	[1.77]	[-0.78]	[3.53]	[2.34]	[4.06]
α_{FF3}	-0.16	0.09	0.04	0.21	0.11	0.27
	[-2.78]	[1.82]	[0.67]	[4.01]	[2.01]	[3.99]
α_{FF4}	-0.17	0.11	0.09	0.16	0.09	0.25
	[-2.85]	[2.12]	[1.58]	[3.10]	[1.50]	[3.63]
α_{FF5}	-0.15	-0.00	0.11	0.10	0.04	0.19
	[-2.59]	[-0.05]	[1.74]	[2.08]	[0.69]	[2.79]
α_{FF6}	-0.16	0.02	0.14	0.08	0.02	0.18
	[-2.67]	[0.36]	[2.36]	[1.52]	[0.43]	[2.64]
Panel B: Fama and French (2018) 6-factor model loadings for LLB-sorted portfolios						
β_{MKT}	1.00	0.94	0.99	0.97	0.97	-0.03
	[72.69]	[81.44]	[70.09]	[82.93]	[73.47]	[-1.70]
β_{SMB}	0.08	-0.09	0.00	0.00	0.08	0.00
	[3.63]	[-5.09]	[0.01]	[0.19]	[3.97]	[0.16]
β_{HML}	0.05	-0.04	-0.21	-0.15	-0.05	-0.09
	[1.73]	[-1.70]	[-7.79]	[-6.81]	[-1.89]	[-3.04]
β_{RMW}	0.00	0.18	-0.04	0.11	0.07	0.06
	[0.16]	[7.81]	[-1.46]	[4.87]	[2.64]	[2.03]
β_{CMA}	-0.04	0.14	-0.17	0.23	0.20	0.24
	[-0.94]	[4.07]	[-4.20]	[6.88]	[5.25]	[5.12]
β_{UMD}	0.01	-0.03	-0.06	0.05	0.02	0.01
	[0.74]	[-2.87]	[-4.26]	[3.90]	[1.74]	[0.80]
Panel C: Average number of firms (n) and market capitalization (me)						
n	546	519	992	562	524	
me ($\$10^6$)	1169	2553	2053	2693	1194	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the LLB strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.24	0.28	0.27	0.25	0.19	0.18
			[3.46]	[4.06]	[3.99]	[3.63]	[2.79]	[2.64]
Quintile	NYSE	EW	0.22	0.24	0.23	0.21	0.22	0.21
			[4.42]	[4.86]	[4.62]	[4.19]	[4.36]	[4.11]
Quintile	Name	VW	0.17	0.21	0.21	0.17	0.13	0.11
			[2.68]	[3.20]	[3.16]	[2.62]	[1.96]	[1.67]
Quintile	Cap	VW	0.20	0.23	0.24	0.20	0.19	0.16
			[3.05]	[3.37]	[3.54]	[2.87]	[2.78]	[2.34]
Decile	NYSE	VW	0.25	0.33	0.29	0.26	0.15	0.14
			[2.82]	[3.71]	[3.26]	[2.93]	[1.68]	[1.60]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.18	0.22	0.21	0.20	0.15	0.14
			[2.63]	[3.23]	[3.14]	[2.95]	[2.17]	[2.03]
Quintile	NYSE	EW	-0.03					
			[-0.51]					
Quintile	Name	VW	0.12	0.15	0.14	0.12	0.08	0.07
			[1.83]	[2.30]	[2.22]	[1.94]	[1.23]	[1.01]
Quintile	Cap	VW	0.15	0.17	0.18	0.16	0.14	0.12
			[2.29]	[2.56]	[2.69]	[2.34]	[2.06]	[1.77]
Decile	NYSE	VW	0.19	0.26	0.22	0.21	0.10	0.09
			[2.07]	[2.89]	[2.48]	[2.32]	[1.13]	[1.07]

Table 3: Conditional sort on size and LLB

This table presents results for conditional double sorts on size and LLB. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on LLB. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high LLB and short stocks with low LLB .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results

	LLB Quintiles					LLB Strategies					
Size quintiles	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
(1)	0.63	0.85	0.95	0.94	0.84	0.21	0.23	0.21	0.15	0.18	0.15
	[2.27]	[2.96]	[3.42]	[3.22]	[2.86]	[1.89]	[2.10]	[1.90]	[1.38]	[1.64]	[1.30]
(2)	0.71	0.92	0.80	0.86	0.89	0.18	0.22	0.17	0.18	0.15	0.16
	[2.71]	[3.52]	[3.09]	[3.37]	[3.56]	[2.23]	[2.60]	[2.16]	[2.23]	[1.80]	[1.91]
(3)	0.74	0.82	0.79	0.89	0.85	0.11	0.14	0.14	0.11	0.12	0.10
	[3.11]	[3.54]	[3.17]	[3.76]	[3.71]	[1.42]	[1.85]	[1.80]	[1.39]	[1.46]	[1.21]
(4)	0.68	0.84	0.81	0.84	0.85	0.17	0.20	0.20	0.19	0.14	0.14
	[3.14]	[3.84]	[3.54]	[3.78]	[4.13]	[2.18]	[2.59]	[2.51]	[2.37]	[1.73]	[1.70]
(5)	0.50	0.62	0.59	0.75	0.70	0.19	0.21	0.21	0.18	0.12	0.10
	[2.77]	[3.52]	[2.89]	[4.08]	[3.88]	[2.19]	[2.34]	[2.35]	[1.99]	[1.27]	[1.10]

Panel B: Portfolio average number of firms and market capitalization

	LLB Quintiles Average n					LLB Quintiles Average market capitalization ($\$10^6$)				
Size quintiles	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
(1)	354	356	355	356	351	32	28	28	27	31
(2)	96	96	96	96	96	52	53	52	52	52
(3)	68	69	68	68	68	92	92	90	91	92
(4)	58	58	58	58	57	201	209	201	208	199
(5)	52	52	52	52	52	1117	1823	1659	1956	1226

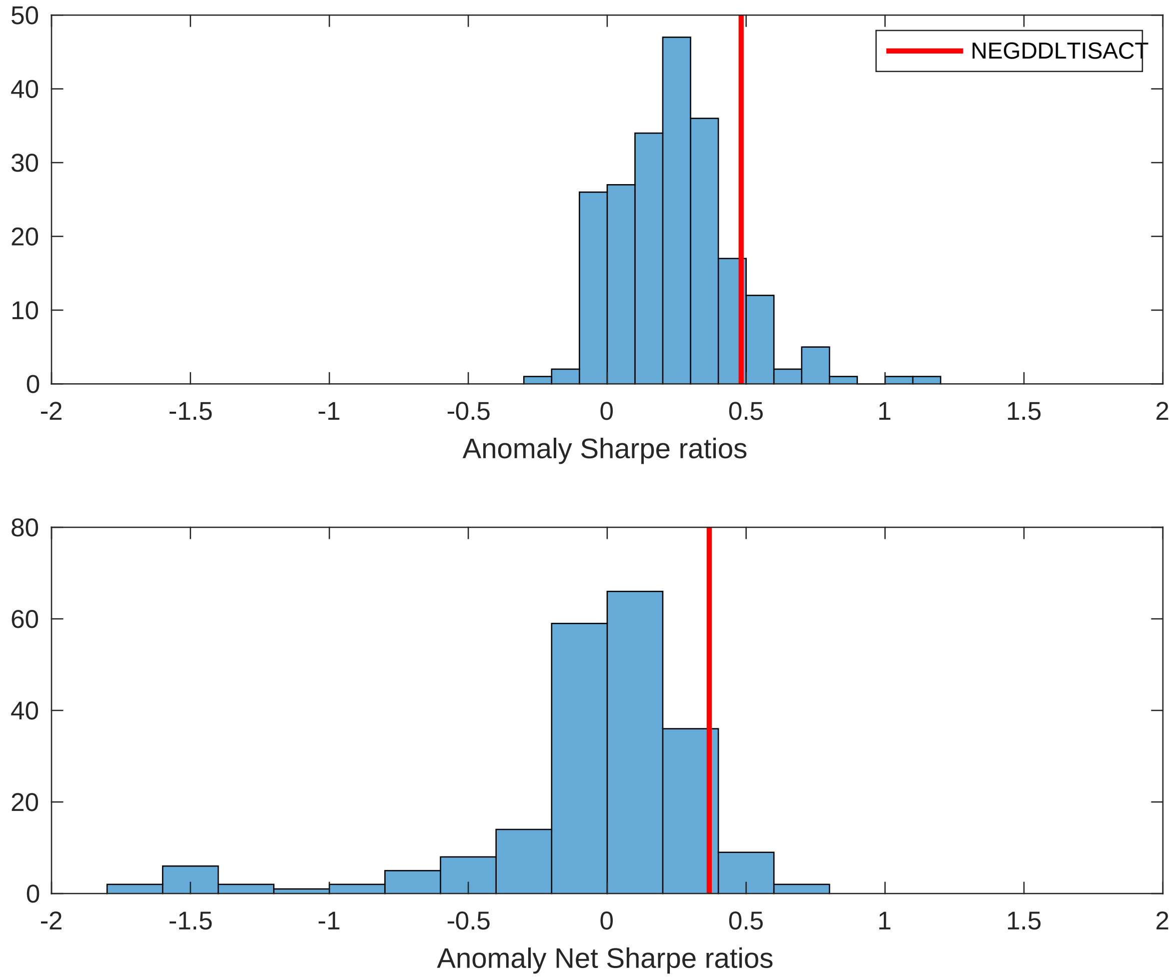


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the LLB with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

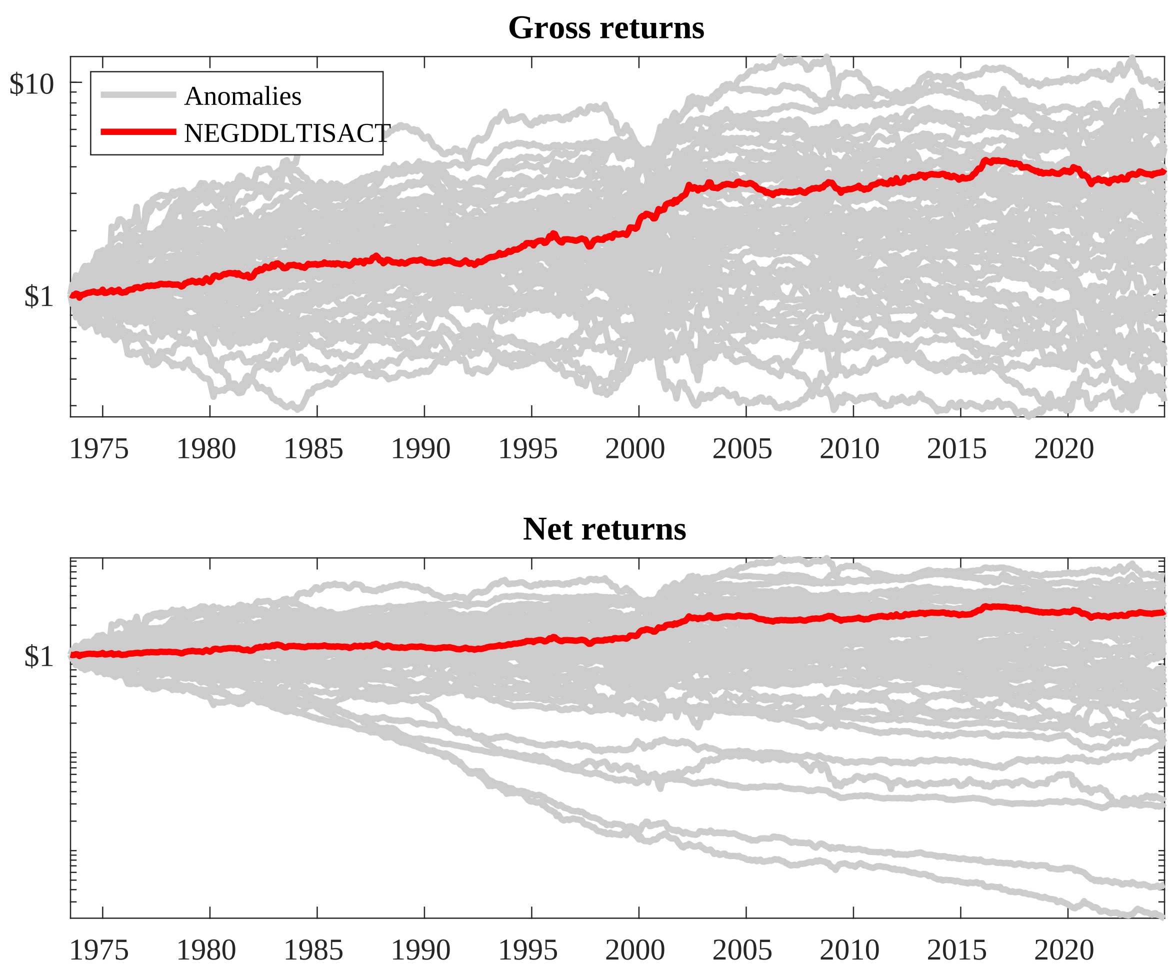
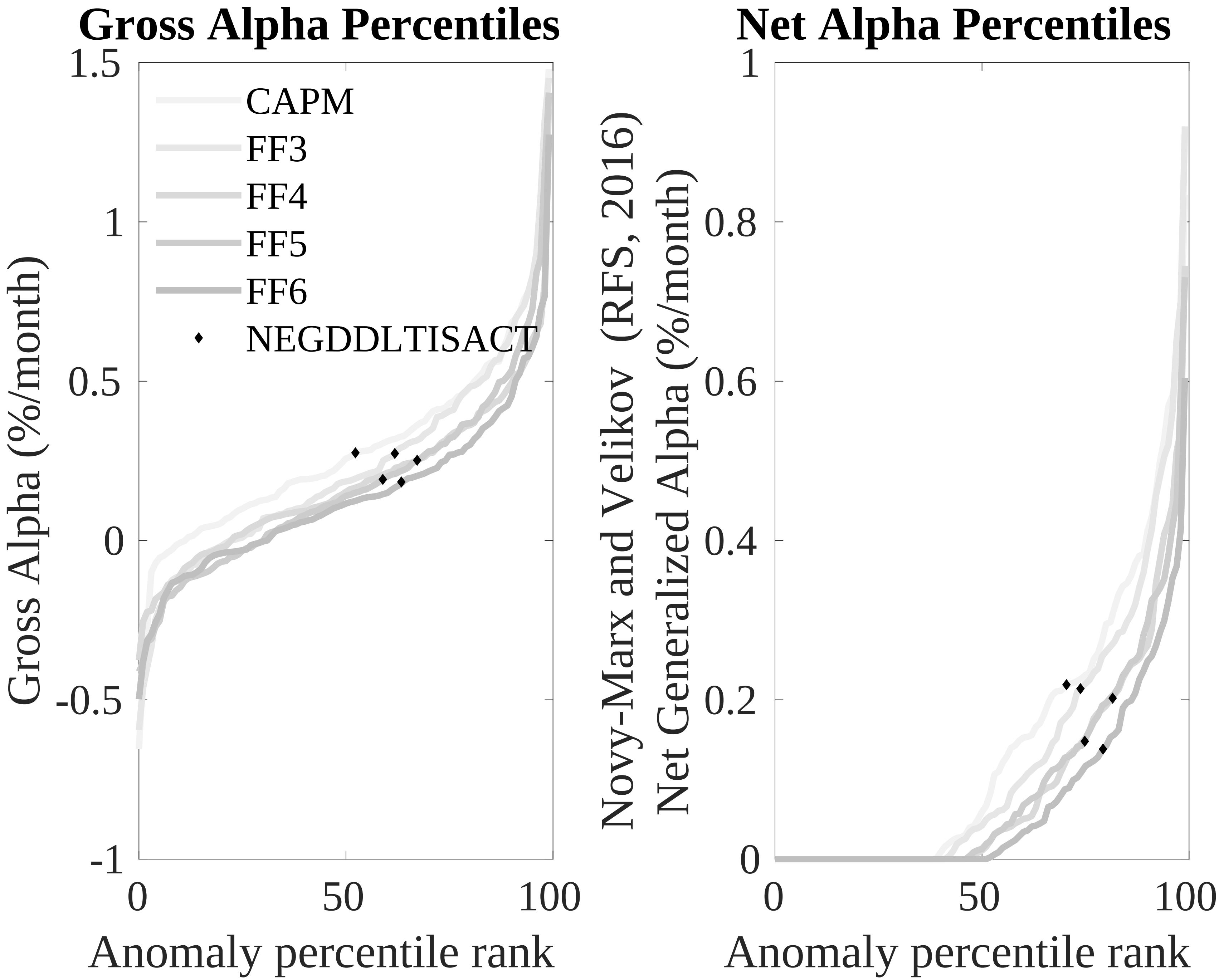


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the LLB trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
CAPM
FF3
FF4
FF5
FF6
NEGDDLTISACT
1.5
1
0.5
0
-0.5
-1
0
50
100
1
0.8
0.6
0.4
0.2
0
0
50
100

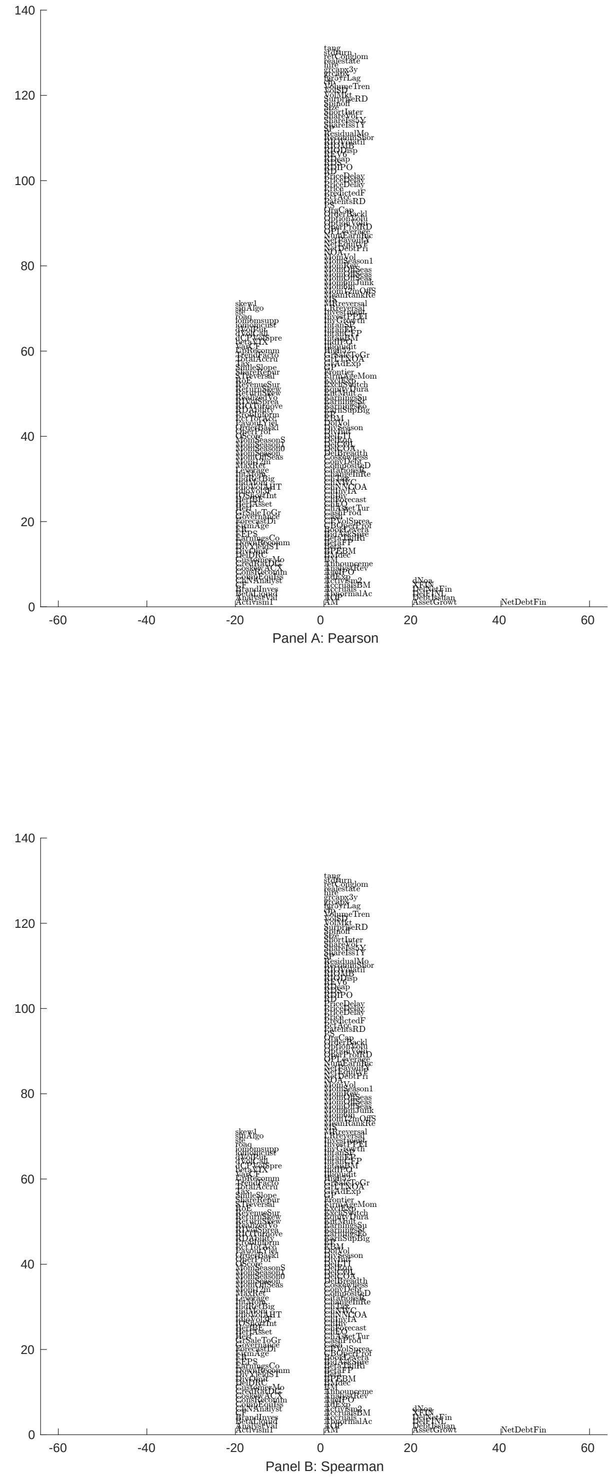


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 209 filtered anomaly signals with LLB. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

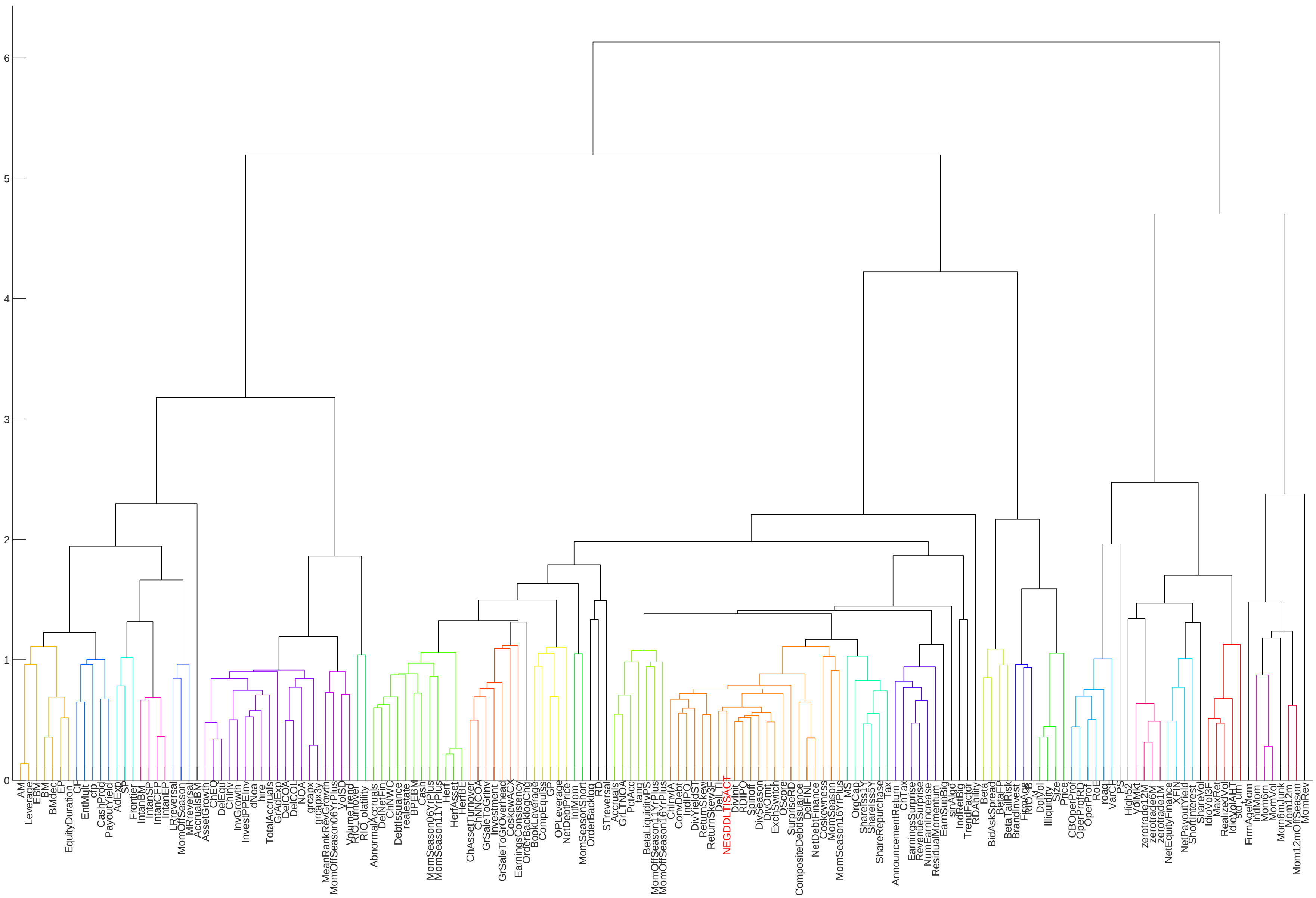


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

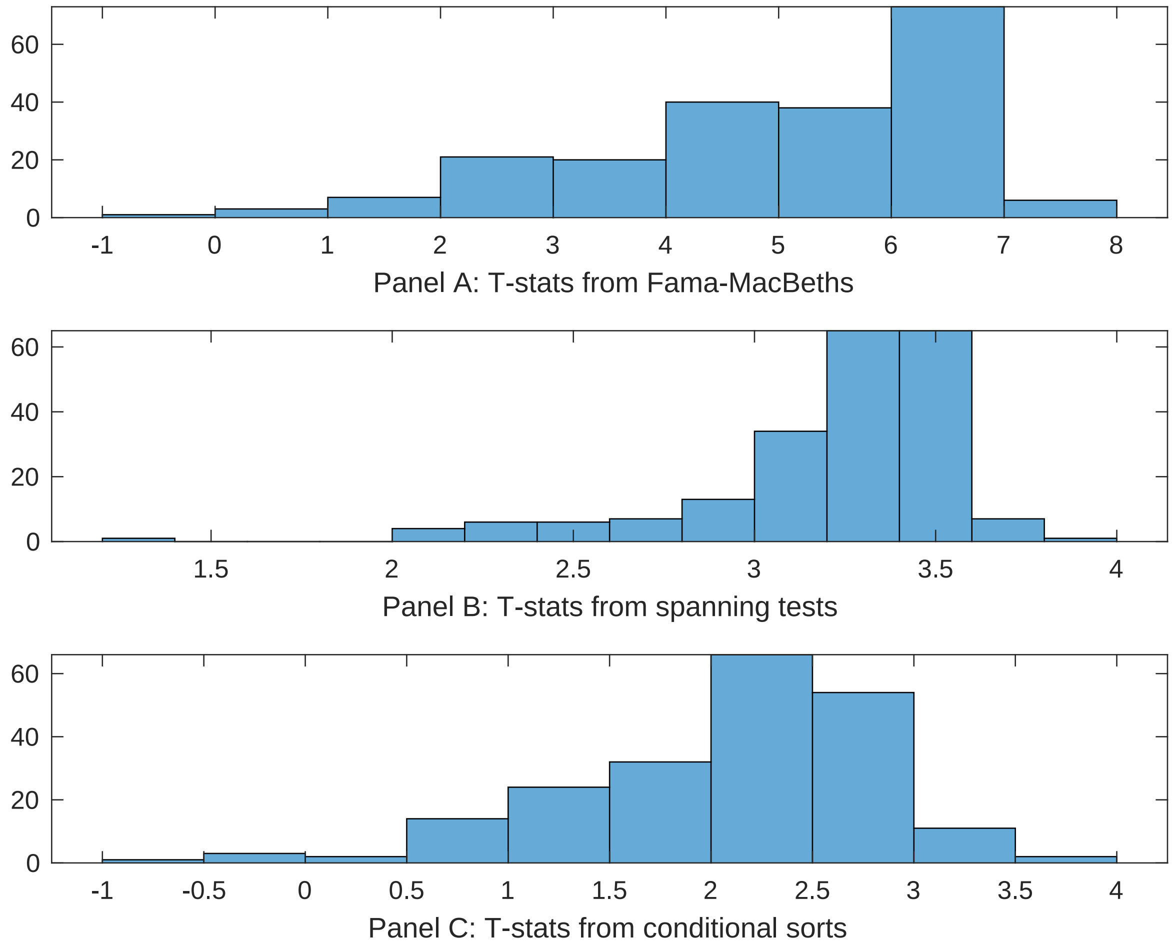


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of LLB conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{LLB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{LLB} LLB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{LLB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on LLB. Stocks are finally grouped into five LLB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted LLB trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on LLB. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{LLB} LLB_{i,t} + \sum_{k=1}^{s} ix \beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Net external financing, Net debt financing, change in ppe and inv/assets, Change in financial liabilities, Inventory Growth, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.14	0.13	0.14	0.13	0.13	0.14	0.15
	[5.66]	[5.24]	[5.69]	[5.28]	[5.30]	[5.72]	[5.75]
LLB	0.80	0.55	0.53	0.41	0.17	0.28	-0.93
	[2.54]	[1.85]	[2.08]	[1.44]	[4.84]	[0.96]	[-0.25]
Anomaly 1	0.19						0.11
	[6.23]						[1.84]
Anomaly 2		0.19					0.97
		[8.46]					[1.45]
Anomaly 3			0.17				0.90
			[7.74]				[3.19]
Anomaly 4				0.17			-0.75
				[9.17]			[-1.57]
Anomaly 5					0.41		0.18
					[6.88]		[0.30]
Anomaly 6						0.11	0.51
						[8.51]	[2.37]
# months	612	612	612	612	612	612	612
$\bar{R}^2$(%)	1	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the LLB trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{LLB} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Net external financing, Net debt financing, change in ppe and inv/assets, Change in financial liabilities, Inventory Growth, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.16	0.17	0.18	0.17	0.18	0.18	0.16
	[2.35]	[2.45]	[2.60]	[2.43]	[2.64]	[2.56]	[2.33]
Anomaly 1	13.63						11.22
	[3.97]						[2.96]
Anomaly 2		12.12					5.95
		[3.22]					[1.15]
Anomaly 3			7.94				3.17
			[2.55]				[0.95]
Anomaly 4				9.04			3.34
				[2.31]			[0.64]
Anomaly 5					6.09		4.63
					[2.26]		[1.63]
Anomaly 6						4.02	-2.12
						[1.00]	[-0.51]
mkt	-0.56	-2.41	-2.54	-2.23	-2.54	-2.44	-0.91
	[-0.34]	[-1.52]	[-1.60]	[-1.40]	[-1.60]	[-1.53]	[-0.55]
smb	5.01	-0.44	0.66	-0.29	1.30	0.34	4.03
	[1.91]	[-0.18]	[0.27]	[-0.12]	[0.54]	[0.14]	[1.44]
hml	-7.64	-8.66	-10.01	-8.55	-9.35	-9.33	-7.87
	[-2.50]	[-2.83]	[-3.25]	[-2.78]	[-3.05]	[-3.03]	[-2.53]
rmw	-1.76	5.06	6.54	5.58	7.27	6.41	-0.55
	[-0.47]	[1.62]	[2.10]	[1.78]	[2.32]	[2.05]	[-0.14]
cma	15.66	21.39	18.26	21.50	19.12	19.66	10.54
	[3.05]	[4.51]	[3.47]	[4.45]	[3.66]	[2.92]	[1.51]
umd	1.39	0.70	1.46	0.70	0.86	1.57	0.19
	[0.89]	[0.44]	[0.93]	[0.44]	[0.54]	[0.99]	[0.12]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	9	8	8	8	8	7	10

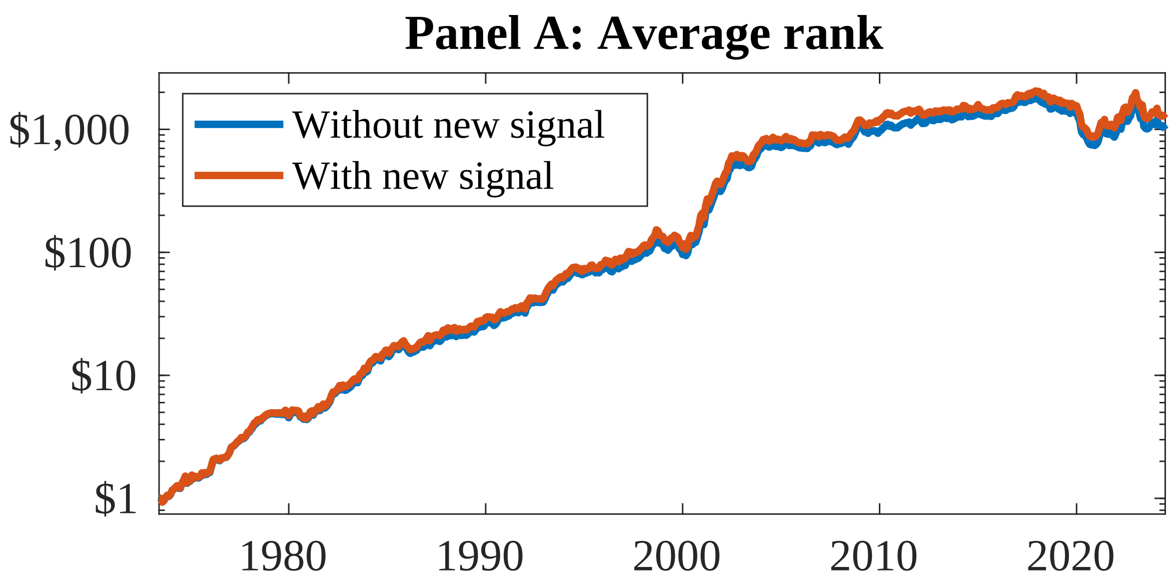


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as LLB. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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